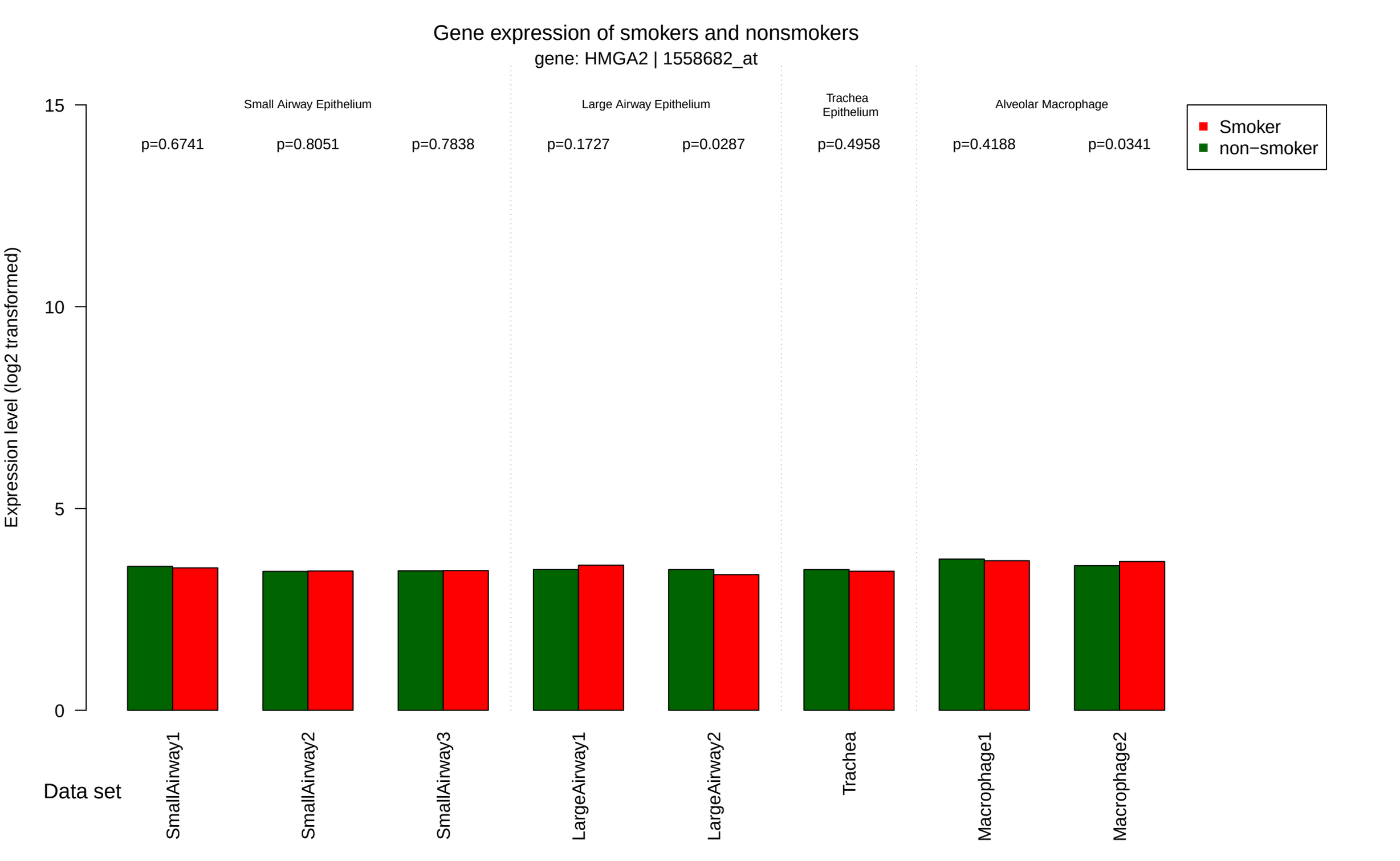
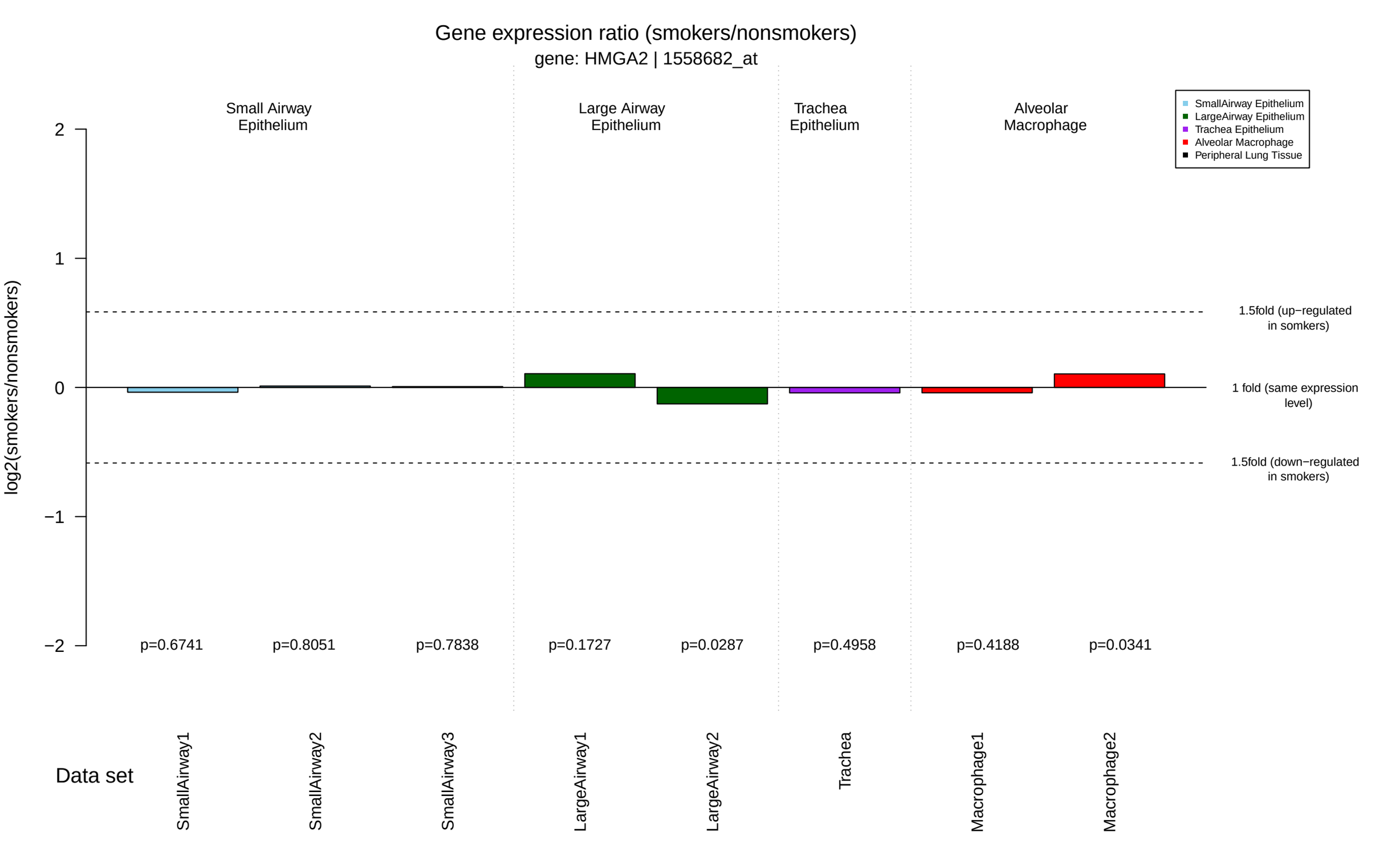
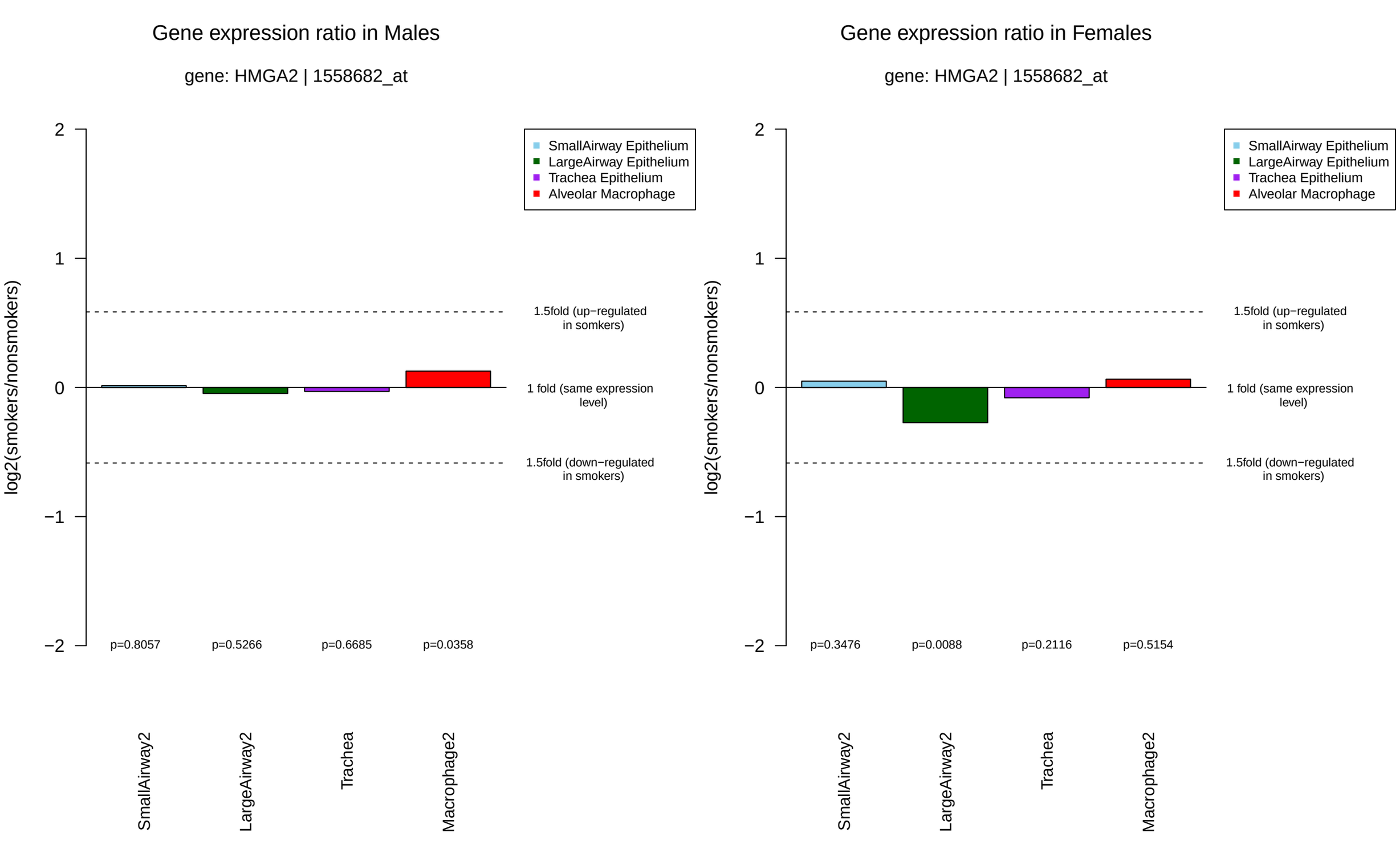
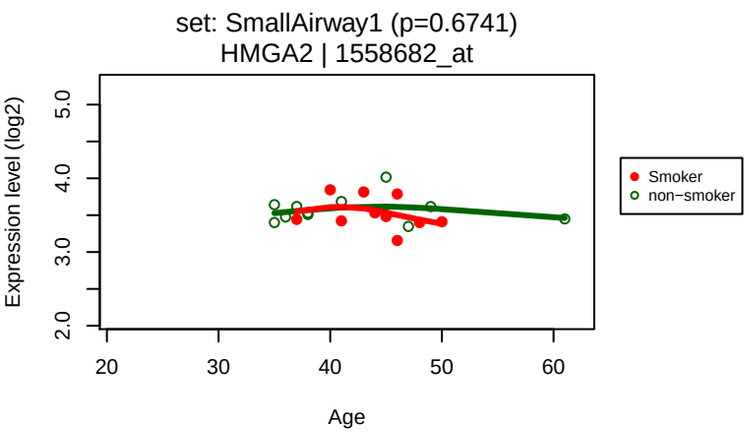
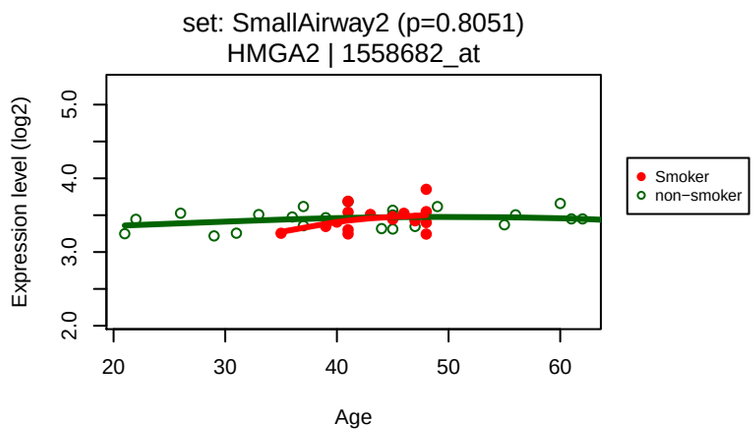
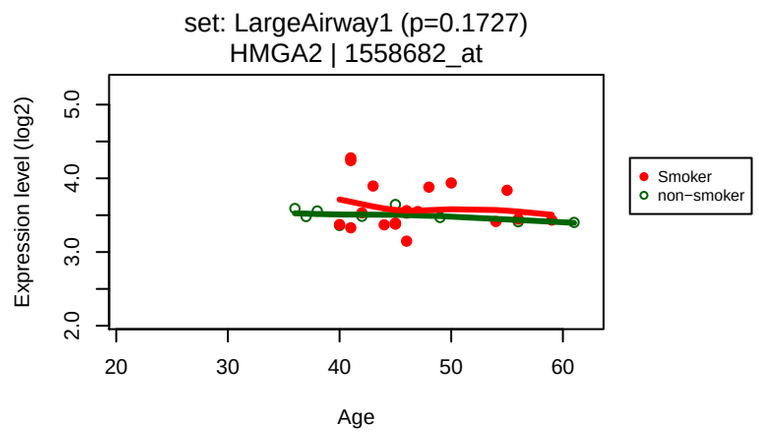

Gene expression of smokers and nonsmokers
gene: HMGA2 | 1558682_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6741
p=0.8051
p=0.7838
p=0.1727
p=0.0287
p=0.4958
p=0.4188
p=0.0341
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

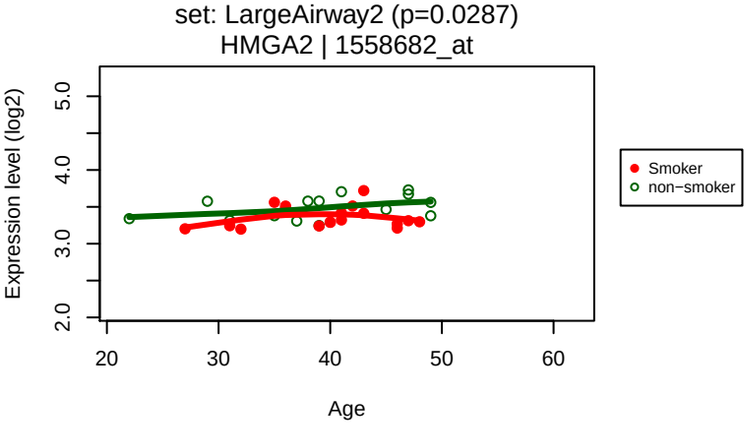

Gene expression ratio (smokers/nonsmokers)
gene: HMGA2 | 1558682_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6741
p=0.8051
p=0.7838
p=0.1727
p=0.0287
p=0.4958
p=0.4188
p=0.0341
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

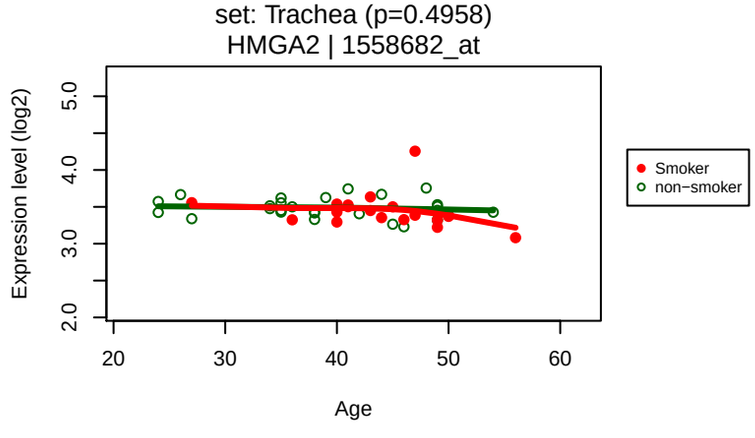

Gene expression ratio in Males
gene: HMGA2 | 1558682_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8057
p=0.5266
p=0.6685
p=0.0358
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGA2 | 1558682_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3476
p=0.0088
p=0.2116
p=0.5154
SmallAirway2
LargeAirway2
Trachea
Macrophage2

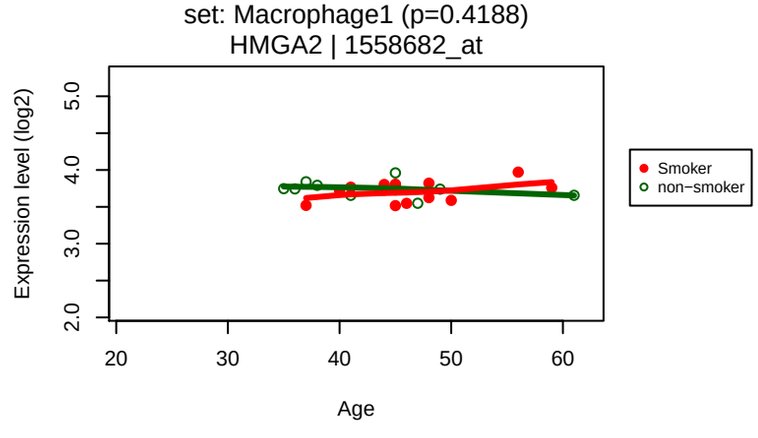

set: SmallAirway1 (p=0.6741)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker

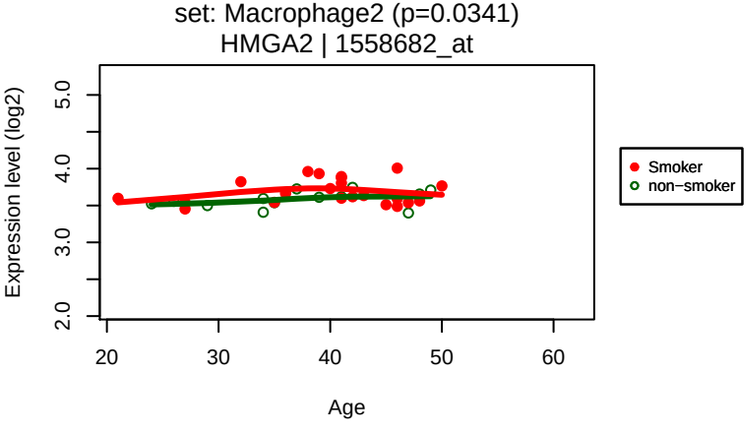

set: SmallAirway2 (p=0.8051)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker

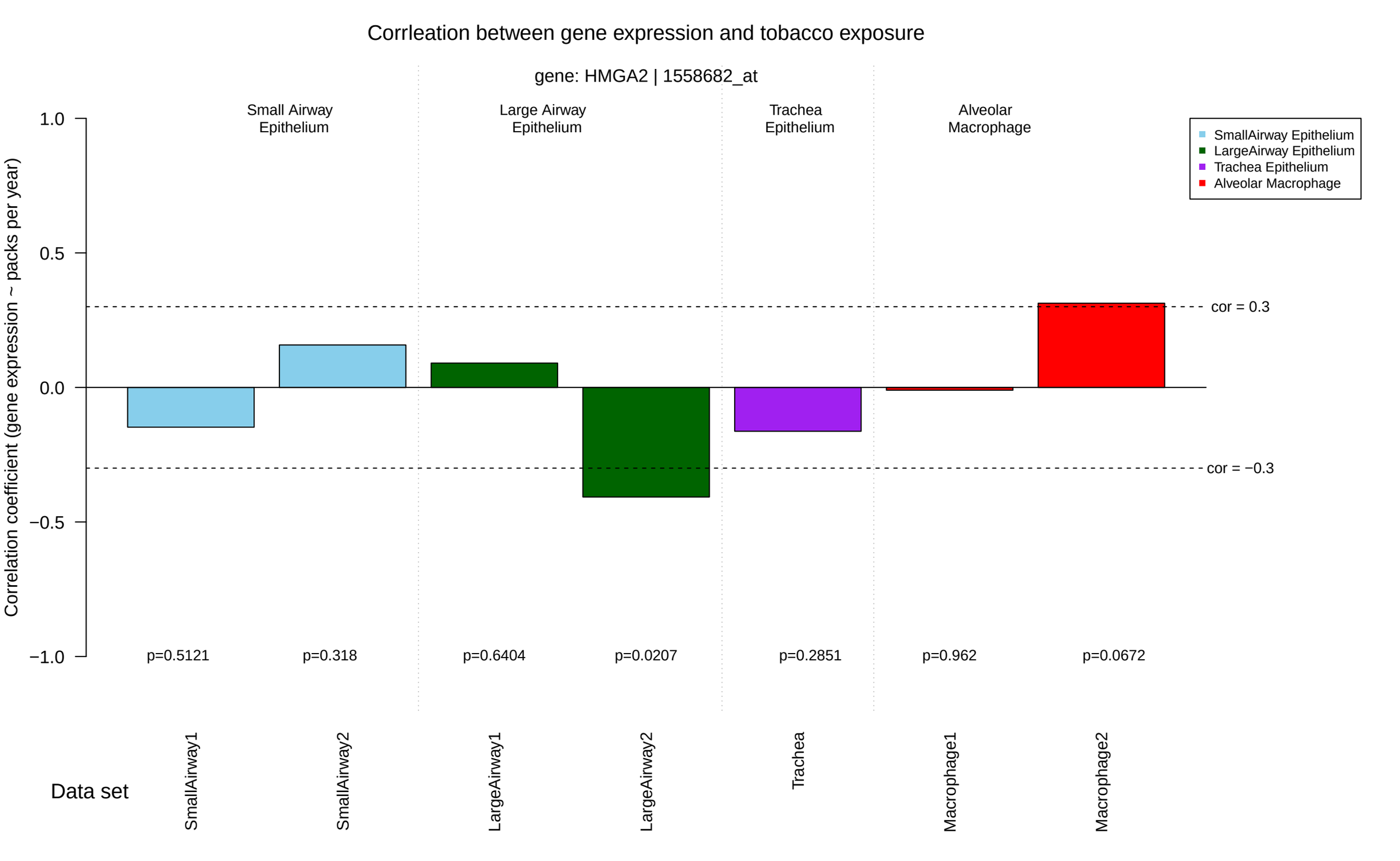

set: LargeAirway1 (p=0.1727)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0287)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4958)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4188)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0341)
HMGA2 | 1558682_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGA2 | 1558682_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5121
p=0.318
p=0.6404
p=0.0207
p=0.2851
p=0.962
p=0.0672
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2